

Arun Prajapat

Software Engineer & · AI/ML Engineer

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EDUCATION

Indian Institute of Information Technology Vadodara

B.Tech in Computer Science and Engineering

August 2023 – May 2027

CPI: 8.4/10

- **Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, Software Engineering, COA, Operating Systems, Computer Networks, Machine Learning, Image Processing, Artificial Intelligence

Vidhya Shri Public School, Didwana (RAJ)

Class XII (RBSE) – Science Stream

July 2021 - April 2022

Percentage: 94.8%

EXPERIENCE

Machine Learning Intern at Infosys Springboard

Dec 2025 - Jan 2026

- Architected a high-performance real-time full-stack web application CryptoPulse for cryptocurrency price analysis and forecasting using React, Python, FastAPI, MongoDB and Binance API.
- Achieved **91-99% accuracy** using **LSTM** and **Gradient Boosting Algorithms** across multiple coins price data.
- Supports 30+ currencies with their real-time historical and future price analysis with 10+ technical indicators.

PROJECTS

AI Research Assistant | Python, LangChain, LangGraph, LangSmith Streamlit, FAISS

March 2026

- Developed an end to end AI Research Assistant chatbot that processes PDFs and generates structural insights.
- Improved response accuracy and contextual relevance by implementing an Agentic Workflow to dynamically handle search, summarization, and citation generation.
- Implemented a **RAG** pipeline by chunking documents, generating embeddings, and retrieving relevant context using vector databases (FAISS) for accurate LLM-based responses.

Dynamic Ensemble for Plant Disease Detection | Python, TensorFlow, NumPy

January 2026

- Developed **Mixture of Experts** deep learning system to detect plant leaf diseases with more than **97% accuracy**.
- Engineered a custom **Gate Network (CNN)** that dynamically assigns weights to **VGG16, VGG19, InceptionV3**, and **ResNet101** predictions based on input image patterns.
- Integrated **Explainable AI (LIME)** to highlight disease-relevant leaf regions, ensuring model transparency and trust for real-world agricultural applications.

Postal Automation | Streamlit, Python, OCR, MongoDB

Oct 2025 – Dec 2025

- Designed an end-to-end automation pipeline (Image Acquisition → Preprocessing → DAB Detection → OCR → Entity Validation → Auto-Correction) to extract destination address from postcard images.
- Implemented a custom multilingual **OCR** engine using **R-CNN Network & CTC Loss** for handwritten text recognition for City name and Pincodes with average **98% accuracy** across all language models.
- Integrated a knowledge graph-based validation module using **MongoDB** to increase accuracy for PIN and City.

MovieFlix : Movie Recommender System | Python, Scikit-learn, NLP, Next.js, MongoDB

August 2025

- Developed a hybrid movie recommendation engine combining Bag of Words (BoW) and Word2Vec embeddings to model both lexical and semantic similarities, improving recommendation relevance using cosine similarity.
- Built a full-stack web application integrating the TMDb API to fetch real-time movie data, enabling an interactive UI for searching and exploring personalized recommendations.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL

Backend & Systems: Node.js, Express.js, REST APIs, FastAPI, WebSockets, JWT

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Databases: MongoDB, MySQL

ML & AI: PyTorch, TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy, LangChain, LangGraph

Tools: Git, GitHub, Postman, VS Code

ACHIEVEMENTS

GATE 2026: GATE 2026 qualified with score 360 in CS branch

Certifications: Infosys Springboard's ML, DL, NLP Developer, NVIDIA Deep Learning.

Coding Profiles: LeetCode Rating (1580) · CodeChef 2-Star (1425)